FENI UNIVERSITY

Center for Learning and Development

Illusionary Number Guessing in Python Turtle Graphics

Course Title	Computer Graphics Lab
Course Code	CSE-328

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Theory

The *Illusionary Number Guess Simulation* is a Python program developed using the **Turtle Graphics** library. It generates a circular illusion pattern and camouflages a random number within it. The player must identify the hidden number from a set of options.

Objectives

- To create a visual illusion using overlapping circles.
- To embed a hidden number within the illusion using camouflage colors.
- To test the player's observation skills by providing multiple-choice options.

Materials & Tools

- **Programming Language:** Python
- **IDE:** Visual Studio Code
- **Library:** Turtle graphics
- Random module for generating random numbers

Algorithm

- **Circular Illusion**: The program draws 250 overlapping circles, each rotated slightly, forming a dense circular illusion.

- **Hidden Number**: A random three-digit number is generated.

It is placed at a random position within the illusion using a camouflage color (lightgray).

- **Options Generation**: The correct number and two random distractors are shuffled into a list of options.

- **User Interaction**: The program prints the options in the console.

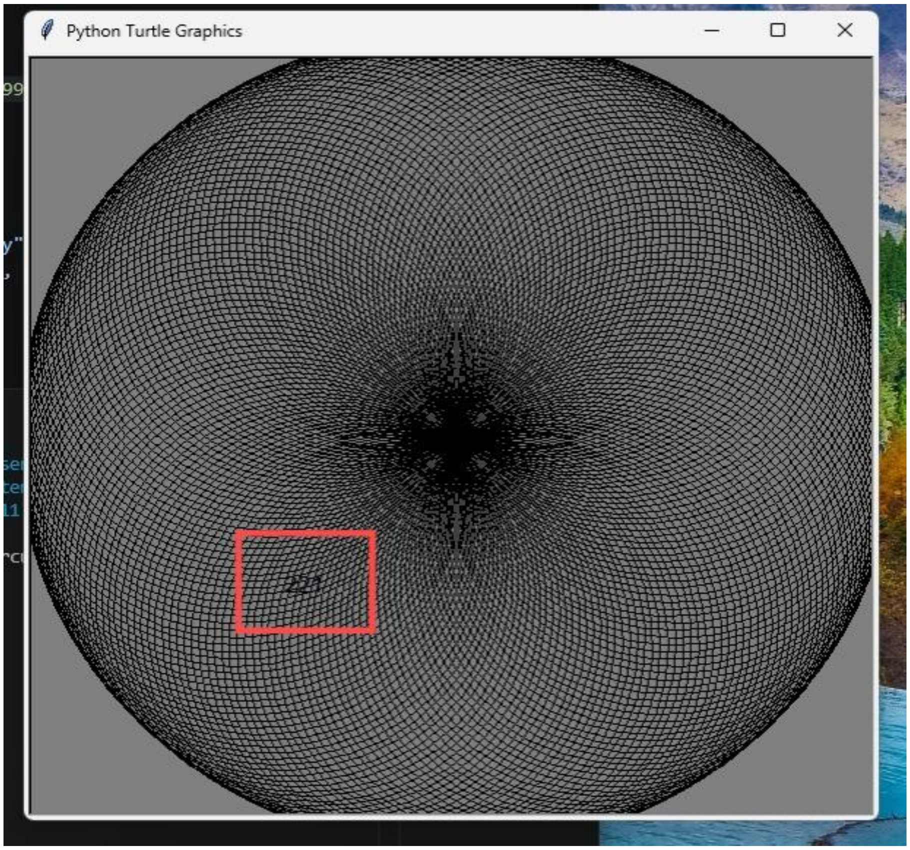
The player enters their guess.

The program checks correctness and displays feedback.

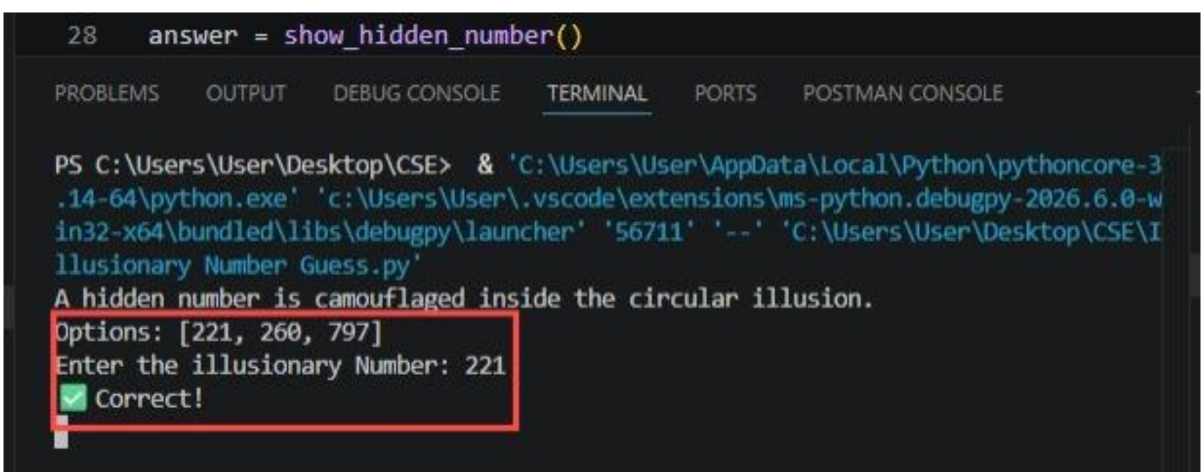
Code Implementation

```
import turtle
import random

screen = turtle.Screen()
screen.bgcolor("gray")
t = turtle.Turtle()
t.speed(0)
turtle.tracer(0)

def circular_illusion():
    t.pencolor("black")
    for i in range(250):
        t.circle(150)
        t.left(2)

def show_hidden_number():
    number = random.randint(100, 999)
    t.penup()
    x = random.randint(-120, 120)
    y = random.randint(-120, 120)
    t.goto(x, y)
    t.pendown()
    camouflage_colors = ["lightgray"]
    t.write(number, align="center", font=("Arial", 12))
    return number

circular_illusion()
answer = show_hidden_number()

choices = [answer, random.randint(100,999), random.randint(100,999)]
random.shuffle(choices)

print("A hidden number is camouflaged inside the circular illusion.")
print("Options:", choices)

guess = int(input("Enter the illusionary Number: "))
if guess == answer:
    print("✅ Correct!")
else:
    print("❌ Wrong! The answer was", answer)

turtle.done()
```

Output/Result



The illusionary number “221” is hidden inside the circular turtle graphics object that is camouflaging the number.



The terminal generated some random numbers as choices to select the correct one. As here, the number 221 is the correct answer.